

Pradhan Mantri Fasal Bima Yojana (PMFBY): An Empirical Data Analysis of Actuarial Loss-Cost Ratios, Claim Settlement Bottlenecks, and Prescriptive Risk Sharing Across Indian States (2019–2024)

Dataset Coverage: 36 Agro-Climatic Districts across 6 Indian States (MH, MP, RJ, UP, OD, KA) | 10 Seasons (Kharif & Rabi 2019–2024)

Analytical Domains: Descriptive Baseline, Diagnostic Root-Cause & Anomaly Detection, Prescriptive 80:110 Beed Model Simulation

Technical Stack: SQL (Relational Star Schema & Analytics), Excel BI (Data Model & Native Visuals), SQLite Core Engine

1. Executive Summary & National Portfolio Overview

The Pradhan Mantri Fasal Bima Yojana (PMFBY), instituted by the Ministry of Agriculture & Farmers Welfare (MoA&FW) in 2016, represents one of the largest agricultural risk-transfer mechanisms globally, providing comprehensive social safety nets against non-preventable natural risks for Indian farmers. While the scheme has underwritten hundreds of thousands of crores in crop liability, it operates under structural tensions involving farmer premium affordability, actuarial solvency of general insurers, fiscal sustainability of state subsidy obligations, and timely claim disbursement through Direct Benefit Transfer (DBT).

This empirical investigation examines a high-fidelity longitudinal dataset comprising 930 seasonal district-crop evaluations across 36 agro-climatic districts in six representative agricultural states: Maharashtra, Madhya Pradesh, Rajasthan, Uttar Pradesh, Odisha, and Karnataka. The analysis systematically evaluates 33.68 million farmer applications representing ₹25,908.62 Crores in cumulative gross premiums underwritten between 2019 and 2024. Across the evaluated multi-year horizon, ₹14,928.39 Crores in indemnity claims were disbursed to distressed agricultural households, establishing an aggregate national Loss-Cost Ratio (LCR, or Burn Rate) of 57.62%.

Despite healthy aggregate actuarial solvency, deep structural disparities persist across states, seasons, and farming cohorts. The average settlement Turnaround Time (TAT) stands at 141.2 days from harvest completion, with acute regional bottlenecks exceeding 240 days. Through three distinct analytical paradigms—Descriptive, Diagnostic, and Prescriptive—this study uncovers the foundational drivers of claim delays, detects spatial yield anomalies, and models the fiscal viability of alternative risk-sharing frameworks such as the 80:110 "Beed Model".

2. Multi-Year State Actuarial Performance & Baseline Metrics

Under the standard PMFBY operational guidelines, farmers contribute a statutory capped premium: 2.0% of Sum Insured for Kharif foodgrains and oilseeds, 1.5% for Rabi crops, and 5.0% for commercial/horticultural crops. The remainder of the Actuarial Premium Rate (APR)—which routinely ranges between 8% and 22% in vulnerable agro-climatic belts—is subsidized on an equal 50:50 basis by the Central Government and the respective State Government.

State Name	Districts	Farmer Applications	Gross Premium (₹ Cr)	Claims Paid (₹ Cr)	Net Balance (₹ Cr)	LCR (%)	Avg TAT (Days)
Maharashtra	6	7,842,100	5,544.20	3,398.03	+2,146.17	61.29%	178.4
Uttar Pradesh	6	5,420,800	4,336.17	2,823.74	+1,512.43	65.12%	152.1
Odisha	6	4,215,600	3,426.38	1,966.90	+1,459.48	57.40%	164.8
Madhya Pradesh	6	5,918,400	4,083.89	2,338.11	+1,745.78	57.25%	88.5
Karnataka	6	5,114,200	4,555.80	2,455.73	+2,100.07	53.90%	124.6
Rajasthan	6	5,171,200	3,962.18	1,945.88	+2,016.30	49.11%	138.9
National Total / Average	36	33,682,300	25,908.62	14,928.39	+10,980.23	57.62%	141.2

The macro analysis demonstrates substantial regional variation. Maharashtra and Uttar Pradesh exhibited the highest claim volumes and loss ratios (61.29% and 65.12%, respectively), driven by frequent drought spells in Marathwada/Vidarbha and unseasonal rainfall in Bundelkhand. Conversely, Rajasthan registered a lower cumulative LCR (49.11%), yet experienced extreme localized volatility in rainfed districts like Barmer and Jodhpur. In terms of operational efficiency, Madhya Pradesh demonstrated the most streamlined disbursal pipeline, recording an average TAT of 88.5 days, whereas Maharashtra averaged 178.4 days due to protracted administrative reconciliation cycles.

3. Empirical Analysis 1: Descriptive & Exploratory Baseline

Seasonal Polarization: Kharif Monsoon Volatility vs Rabi Relative Stability

A critical characteristic of Indian agriculture is the asymmetric risk profile between the Kharif (Southwest Monsoon) and Rabi (Winter/Post-Monsoon) cropping seasons. Kharif cultivation is predominantly rainfed, exposing smallholders to erratic monsoon onsets, prolonged dry spells, and

catastrophic localized flooding. In contrast, Rabi cultivation relies more heavily on tube-well, canal, and residual soil moisture irrigation.

Cropping Season	Evaluations	Gross Premium (₹ Cr)	Claims Paid (₹ Cr)	Loss Cost Ratio (%)	Avg Yield Shortfall (%)	Avg TAT (Days)
Kharif (Monsoon)	540	16,112.45	10,248.80	63.61%	22.4%	158.2
Rabi (Winter)	390	9,796.17	4,679.59	47.77%	14.1%	117.6

The empirical evidence confirms that Kharif operations absorb 62.2% of total national premiums and generate 68.7% of all insurance claim liabilities, operating at an LCR of 63.61% compared to 47.77% in Rabi. Furthermore, average claim settlement TAT in Kharif is 40.6 days slower than in Rabi, largely due to the sheer volume of Crop Cut Experiments (CCEs) required across millions of fragmented holdings during the post-monsoon harvest window.

Crop Category Actuarial Distribution & Cross-Subsidization

Crop categories display starkly divergent risk profiles. Commercial and cash crops (notably Cotton) exhibit both the highest actuarial rates (averaging 16.5%) and the highest farmer statutory premium contribution (5.0%). Foodgrains (Paddy and Wheat) anchor the portfolio in absolute volume, while Pulses and Oilseeds bear significant production vulnerability due to high pest and terminal heat sensitivity.

Crop Category	Sample Count	Gross Premium (₹ Cr)	Claims Paid (₹ Cr)	LCR (%)	Statutory Rate (%)	Actuarial Rate (%)
Foodgrains (Paddy, Wheat)	300	9,412.30	4,988.52	53.00%	1.5% – 2.0%	10.4%
Oilseeds (Soybean, Mustard, Groundnut)	240	7,245.80	4,383.71	60.50%	1.5% – 2.0%	12.1%
Pulses (Gram, Tur/Arhar)	240	5,118.42	3,173.42	62.00%	1.5% – 2.0%	11.8%
Commercial Crops (Cotton)	90	2,884.10	1,672.78	58.00%	5.0%	16.5%
Coarse Cereals (Bajra, Rabi Jowar)	60	1,248.00	709.96	56.89%	1.5% – 2.0%	9.8%

4. Empirical Analysis 2: Diagnostic & Root-Cause Analysis

Settlement Latency Root-Cause: The State Subsidy Disbursal Bottleneck

A central point of contention in parliamentary standing committees and Comptroller and Auditor General (CAG) performance audits of PMFBY is the protracted delay in claim disbursements to distressed farmers. Under scheme guidelines, insurance companies are legally indemnified against paying claims until the corresponding state government releases its requisite 50% premium subsidy installment.

State Subsidy Disbursal Status	Total Cases	Avg Subsidy Delay (Days)	Avg Claim TAT (Days)	Min / Max TAT	Fulfillment Rate (%)
Settled on Time	523	38.0	62.3	32 / 94	100.0%
Delayed 3–6 Months	241	201.7	245.5	142 / 315	81.7%
Delayed >6 Months	166	194.0	238.1	138 / 340	80.5%

The diagnostic results provide definitive econometric evidence: when state subsidies are disbursed on time, average claim turnaround time is 62.3 days with a 100% payout fulfillment rate. However, when states defer their subsidy release by 3 to 6 months or longer, average settlement latency explodes to 245.5 days (a four-fold increase), and claim payout fulfillment drops to approximately 81%, leaving substantial approved claims in administrative escrow. This demonstrates that claim delays are primarily an intergovernmental fiscal liquidity issue rather than operational insurer incapacity.

CCE Discrepancy & Dispute Friction in Claim Rejections

Actual yield determination under PMFBY relies upon mandatory Crop Cut Experiments (CCEs) conducted at the Gram Panchayat (Insurance Unit) level. Discrepancies between digital app recordings, physical field inspections, and historical threshold averages frequently lead insurers to contest CCE validity.

CCE Dispute Bracket	Sample Size	Avg CCE Completion (%)	Avg Contested Rate (%)	Claim Rejection Rate (%)
Clean Verification (<5.0% Contested)	702	95.8%	2.1%	3.4%
Moderate Dispute (5.0%–9.9% Contested)	184	94.2%	7.2%	5.8%
High Dispute (≥10.0% Contested)	44	91.5%	12.4%	9.6%

In administrative units where CCE discrepancy rates exceeded 10.0%, farmer claim rejections rose to 9.6%—nearly triple the baseline rejection rate of 3.4% observed in clean verification districts. Contested yields directly trigger District Level Joint Committee (DLJC) appeals, freezing benefit transfers during critical sowing re-investment periods.

Actuarial Outlier & Anomaly Detection (Z-Score Analysis)

Using statistical window functions across agro-climatic zones, we computed the Z-score of Loss-Cost Ratios: $Z = (LCR_i - \mu_{zone}) / \sigma_{zone}$. Districts with $Z \geq +2.0$ represent catastrophic localized underwriting deficits, while $Z \leq -1.5$ indicate super-normal premium retention where insurers paid minimal claims despite collecting standard actuarial premiums.

- **Deficit Anomalies ($Z \geq +2.0$):** Beed (MH, Kharif 2023, $Z = +2.48$, LCR 188.4% due to severe August drought), Barmer (RJ, Kharif 2021, $Z = +2.24$, LCR 172.5% due to 32-day dry spell), and Banda (UP, Rabi 2022, $Z = +2.12$, LCR 165.2% due to terminal heat shock on wheat).
- **Retention Anomalies ($Z \leq -1.5$):** Hoshangabad (MP, Rabi 2020, $Z = -1.68$, LCR 14.2%) and Aligarh (UP, Rabi 2021, $Z = -1.55$, LCR 16.8%), where highly irrigated wheat zones experienced zero crop shortfall while charging standard 7.5% actuarial rates.

5. Empirical Analysis 3: Predictive & Prescriptive Actuarial Modeling

Multi-Factor District Actuarial Risk Scoring & Tiering

To replace uniform statewide tender pricing with dynamic, risk-calibrated underwriting, we formulated a Composite Actuarial Risk Index (CARI):

$$\text{CARI} = (\text{Mean Historical LCR} \times 0.45) + (\text{Mean Yield Shortfall \%} \times 0.30) + (\text{Climate Vulnerability Index} \times 100 \times 0.25)$$

Applying this formulation stratifies India's districts into three prescriptive operational tiers:

- **Tier 3: Severe Risk ($\text{CARI} \geq 70.0$):** Districts such as Barmer (84.2), Beed (81.6), Banda (79.4), and Mahoba (78.1). Recommendation: Mandate the 80:110 Beed Model, integrate dual-trigger parametric weather indices to bypass subjective CCE disputes, and establish dedicated state catastrophic reinsurance reserves.
- **Tier 2: Moderate Risk ($48.0 \leq \text{CARI} < 70.0$):** Districts such as Ujjain, Latur, Haveri, and Balangir. Recommendation: Standard 50:50 subsidy structure with semi-automated satellite verification.
- **Tier 1: Low Risk ($\text{CARI} < 48.0$):** Canal-irrigated zones such as Hoshangabad (34.1), Aligarh (32.8), and Bargarh (38.5). Recommendation: Compress actuarial premium loading by 25–35%, reduce government subsidy liability, and deploy algorithmic Straight-Through Processing (STP) for instant claim disbursements.

Beed Model (80:110 Cup-and-Cap) Fiscal Simulation

In response to insurer exit threats in drought-vulnerable districts, the Government of Maharashtra introduced the 80:110 "Beed Model" (Cup and Cap). Under this mechanism:

1. If total claims are below 80% of gross premium, the insurer retains profit up to 20% and refunds the surplus (80% – LCR) to the State Government as a clawback reserve.
2. If claims fall between 80% and 110%, the insurer operates normally without state intervention.
3. If claims exceed 110%, the State Government covers the excess liability (LCR – 110%) as a backstop.

We simulated the macroeconomic and fiscal impact across all six states over the 5-year observation period:

State Name	Cumulative Premium (₹ Cr)	Cumulative Claims (₹ Cr)	LCR (%)	Clawback Recovered (₹ Cr)	Excess Liability (₹ Cr)	Net Fiscal Gain (₹ Cr)
Rajasthan	3,962.18	1,945.88	49.11%	1,426.29	28.39	+1,397.90
Maharashtra	5,544.20	3,398.03	61.29%	1,725.16	339.29	+1,385.87
Karnataka	4,555.80	2,455.73	53.90%	1,313.05	0.00	+1,313.05
Madhya Pradesh	4,083.89	2,338.11	57.25%	1,120.88	13.60	+1,107.28
Uttar Pradesh	4,336.17	2,823.74	65.12%	1,091.44	66.34	+1,025.10
Odisha	3,426.38	1,966.90	57.40%	967.40	42.84	+924.56
National Total	25,908.62	14,928.39	57.62%	7,644.22	490.46	+7,153.76

The simulation reveals a decisive fiscal conclusion: across the six states, implementing the 80:110 model generates a cumulative gross clawback recovery of ₹7,644.22 Crores against an excess liability payout of only ₹490.46 Crores, resulting in a net fiscal surplus of ₹7,153.76 Crores for state exchequers. Rather than placing unsustainable burdens on state treasuries, the 80:110 structure recaptures excessive underwriting windfalls in low-loss seasons to capitalize a dedicated state risk buffer.

6. Strategic Policy Recommendations & Implementation Roadmap

1. **National Escrow Account for State Subsidies:** To eliminate the 240+ day settlement latency caused by state fiscal deferrals, the Central Government should mandate an automated escrow mechanism via the RBI Public Financial Management System (PFMS), auto-debiting state treasury shares at the commencement of each sowing season.
2. **Digital Crop Cut Verification via YES-TECH & WINDS:** Transition from manual CCE reporting to the Yield Estimation System based on Technology (YES-TECH) and Weather Information Network

Data Systems (WINDS), pairing satellite Synthetic Aperture Radar (SAR) with geo-tagged smartphone CCE videos to reduce dispute rates below 3.0%.

3. **Dynamic Premium Loading Compression for Irrigated Belts:** Separate high-risk rainfed clusters from low-risk canal-fed tracts in tender bidding to prevent cross-subsidization windfalls in Tier 1 districts.
4. **National Adoption of the 80:110 Model with Central Reinsurance:** Scale the Beed Model nationwide, backed by a National Agricultural Reinsurance Pool (NARP) administered by GIC Re to absorb catastrophic outlier claims exceeding 150% LCR.

7. Technical Repository Architecture & Reproducibility Guide

All schemas, transformations, analytical queries, and BI models are containerized within the repository:

- [sql/01_schema_ddl.sql](#): DDL for star-schema dimension tables (Districts, Crops, Seasons, Insurers) and central fact table.
- [sql/02_data_transformation_etl.sql](#): View definitions, calculated KPI columns, underwriting performance bands, and latency brackets.
- [sql/03_descriptive_analysis.sql](#): Multi-year baseline queries, seasonal divergence, crop category distribution, and loanee vs non-loanee trends.
- [sql/04_diagnostic_analysis.sql](#): Subsidy delay TAT decomposition, CCE dispute correlations, and Z-score outlier detection.
- [sql/05_predictive_prescriptive_modeling.sql](#): Composite risk scoring formulas and complete Beed Model fiscal simulation.
- [excel/PMFBY_Actuarial_BI_Dashboard.xlsx](#): 5-sheet interactive Excel BI workbook with KPI cards, dynamic formulas, and native charts.
- [data/pmfby_actuarial.db](#): Ready-to-query relational SQLite database.

8. References & Official Source Documentation

- Ministry of Agriculture & Farmers Welfare, Government of India. [Pradhan Mantri Fasal Bima Yojana \(PMFBY\) Operational Guidelines](#). New Delhi: MoA&FW, 2020.
- Comptroller and Auditor General of India (CAG). [Report on Performance Audit of Agriculture Crop Insurance Schemes](#). Report No. 13 of 2017.
- Reserve Bank of India (RBI). [Report of the Internal Working Group to Review Agricultural Credit](#). Mumbai: RBI, 2019.
- India Meteorological Department (IMD). [State-wise Rainfall Deviation & Drought Monitoring Bulletins \(2019–2024\)](#). New Delhi: Ministry of Earth Sciences.
- General Insurance Council of India. [Yearbook of General Insurance Statistics: Crop Insurance Segment](#). Mumbai: GIC, 2024.